ECE 447

Lesson 08
Amplitude Modulation, Part 1



UNITED STATES
AIR FORCE
ACADEMY

Schedule and Admin

- Schedule
- Admin
  - **HW 1 due tonight.** Any issues?
  - **Lab 2 Assignment**. The assignment associated with Lab 2 is due Lesson 10 (Tue, 1 Sep) by 2359 via Gradescope upload.
  - **HW 2 assigned next lesson.** Only two problems - shorter HWs, assigned more often. Don't get behind! HWs and Labs reinforce the lesson material.

Did You Do the Reading?

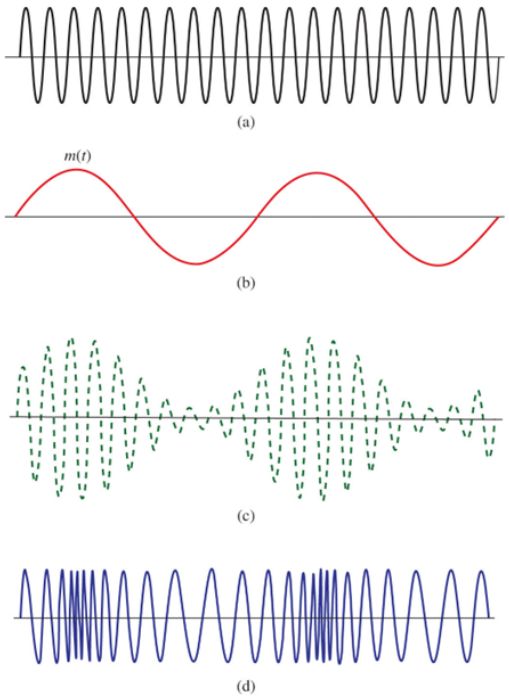
Where do we see *baseband communications*?

Where do we see *carrier communications*?

Modulation means *changing* something about a signal - and not always just shifting the signal's frequencies

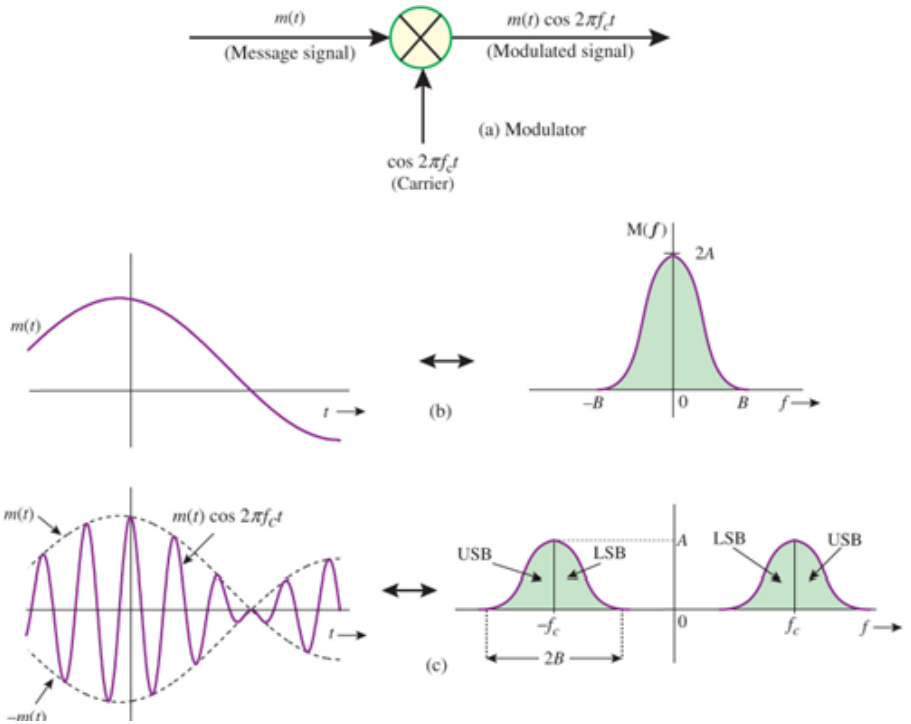
Modulation

- Message signal: $m(t)$
- General carrier modulated signal: $s(t) = A(t)\cos\left[2\pi f_c t + \phi(t)\right]$
- Amplitude (linear) modulation, $A(t)$
  - $s_1(t) = m(t) \cdot \cos\left(2\pi f_c t\right)$
  - $S_1(f) = \frac{1}{2}M(f - f_c) + \frac{1}{2}M(f + f_c)$
- Angular modulation, f_c or $\phi(t)$
  - Message modulates frequency or phase
  - Notice the general equation doesn't have $f_c(t)$ - FM is a little different - more to come in the angle modulation lessons!



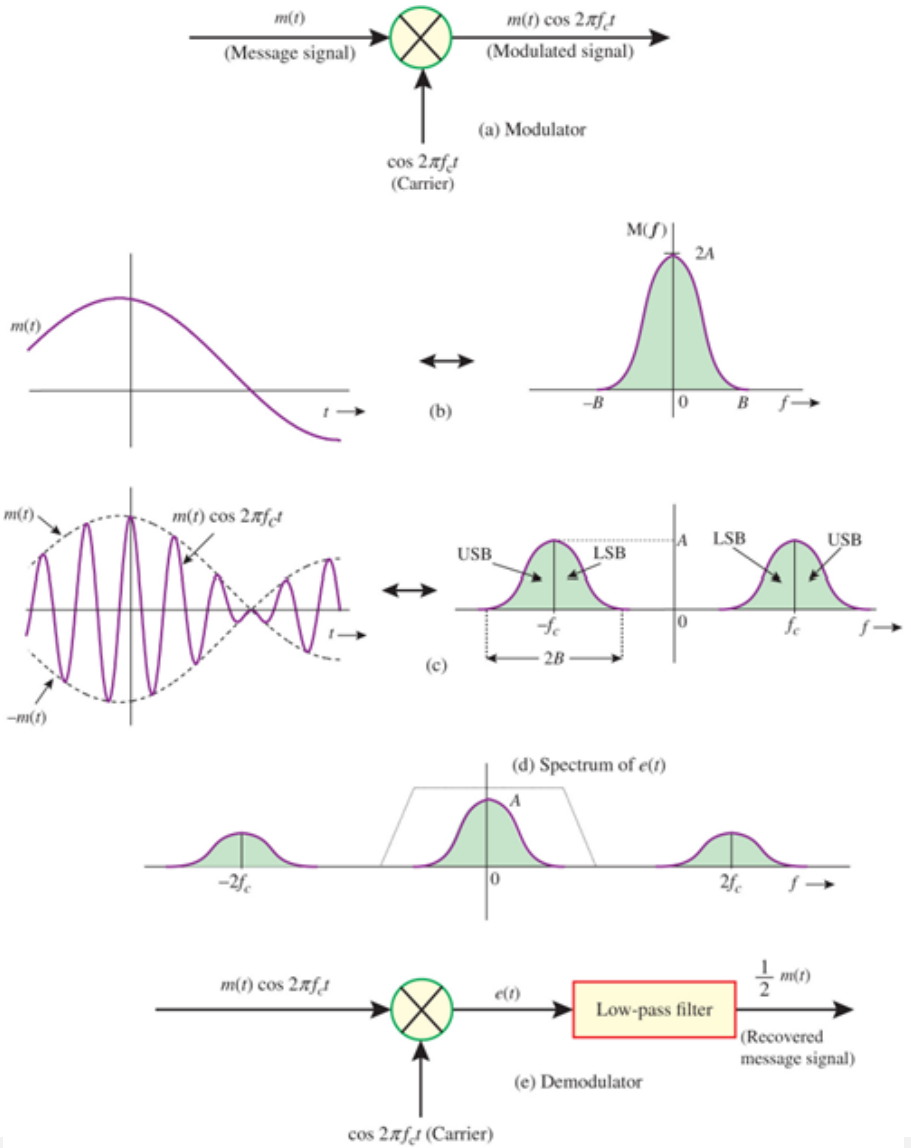
DOUBLE-SIDEBAND, SUPPRESSED-CARRIER (DSB-SC)

- Simplest form of AM
- Learned this in ECE 215
- $\varphi_{DSB-SC}(t) = A_c m(t)\cos(\omega_c t)$
- Over-, under-, or 100%-modulated?



Double-Sideband, Suppressed-Carrier (DSB-SC)

- Demodulation
  - Same as modulation with LPF
  - What if $f_c < B$?
  - Reality, $f_c/B \gg 1$ to avoid distortion
  - Skills Review, Prob 6 - what happened when the receiver didn't have the exact frequency/phase?
  - Difficult to achieve synchronized carrier signals for receiver



Extra Time?

Free EI!